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Task Description:	We have identified recessive mutations at three loci defective in the signal transduction events of gravitropism in Arabidopsis. The gravity persistent signal (gps) mutants were isolated using a temperature effect on gravity signal transduction and response. gps1 did not bend in response to the 4oC gravity stimulation upon return to 21oC. gps 2 responded to the 4oC gravity stimulus with the same kinetics, but in the opposite direction as wild type. gps 3 mutants over-responded after return to room temperature, continuing to bend past the predicted 90o reorientation. The gps mutants are some of the first mutants to be reported that are affected in aspects of gravitropic signal transduction prior to auxin transport, including how polarity is established. We have cloned the genes defective in all three mutants and are now characterizing their expression in wild-type plants so that we can better understand their role in gravitropism. In addition, we have identified several potential new mutants in the GPS mechanism. The objective of the proposed research was to identify components of the gravity signal transduction pathway. We have identified a group of mutants		

	specifically defective in the events of signal transduction and the specific objectives of this research were to characterize the mutants, clone the genes affected, and identify new mutants in the pathway. The research should provide us with greater insight into the molecular components of gravity signal transduction, an area that remains a “black box” in plant gravitropism.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gravitropism is an important developmental process that allows plant organs to use gravity as a guide for growth. In inflorescence stems, gravity sensing seems to involve the sedimentation of amyloplasts in the endodermis. Following the displacement of amyloplasts upon gravistimulation, lateral redistribution of auxin is proposed to initiate differential growth along the elongation zone that results in curvature. The primary objective of this proposal is to improve our understanding of the signal transmission phase of gravitropism through the study of a new group of mutants defective in the early signal transduction events that link amyloplast sedimentation to auxin redistribution. The proposed experiments should provide data allowing a better understanding of the signal transduction events that establish auxin redistribution during gravitropism. The identification of genes involved in the GPS response should enhance our knowledge of the components of the signal transduction pathway leading from gravity stimulus perception to the differential growth response of inflorescence stems. To date, little is known about this important aspect of plant gravitational biology. Gravitropism allows plants to orient their organs optimally for photosynthesis or water and nutrient uptake. Besides providing new, fundamental information to supplement existing models of plant gravitropism, elucidating factors governing plant responses to gravitational forces has potentially far-reaching implications for manipulating plant growth and development for human use and for understanding other signal transduction events, especially those initiated by biophysical stimuli. Besides providing new, fundamental information to supplement existing models of plant gravitropism, results of this research could allow advances in applied research that might enable the manipulation of plant organ position, increasing access for harvest, and lead to innovations that will improve plant use for bioregenerative life support for extended spacefight. In addition, polar auxin transport controls multiple aspects of plant growth and development, including tropisms, embryo axis formation, vascular development, lateral root formation and apical dominance. Therefore, a better understanding of the events that establish auxin redistribution during gravitropism could increase our fundamental understanding of the processes involved in plant development.
Task Progress:	The proposed research focused on the genetic and physiological characterization of the gravity persistent signal (gps) mutants of Arabidopsis. The gps mutants were isolated using a chilling effect on gravity signal transduction that separated stimulus perception from response. gps1 mutants did not bend in response to the 4oC gravity stimulation upon return to 21oC. gps 2 responded to the 4oC gravity stimulus by bending in the opposite direction to wild type. gps 3 mutants over-responded after return to room temperature. In the first year of the project, we were able to characterize all three mutants with regard to their physiology, statolith sedimentation and auxin transport. We were also able to clone the genes defective in gps1 and gps3. In the second year, we have performed subcellular localization of these genes, made strides to complement the mutant phenotype and made the necessary constructs toward assessing expression of the gene in vivo using a green fluorescent protein fusion. We have also performed a transcription profiling experiment with gps1 in hope of better understanding the effect of the mutation of expression of other genes involved in gravitropism and have identified a family member of GPS1 that is expressed in root tissue. In addition, we have cloned the gene defective in gps2 and have identified several new potential mutants in the GPS pathway. GPS1 was cloned using Thermal Asymmetric Interlaced Polymerase Chain Reaction (TAIL PCR). The gene identified in both gps1-1 and gps1-4 was CYP705A22, a p450 of unknown function, located on chromosome 3 of Arabidopsis. Building constructs for complementation of the gps1 phenotype has proven problematic, but we now have a construct for the genomic DNA for CYP705A22 under control of both the 35S promoter and the endogenous promoter ready for transformation into gps1 plants. To characterize timing and tissue specificity of gene expression, we have also constructed a variety of green fluorescent protein (GFP) fusions. We have transiently expressed a GFP:CYP705A22 transit peptide fusion in onion cells and BY2 tobacco suspension culture cells providing information on subcellular localization. The gene appears to be targeted to the endomembrane system. In addition, the promoter for the CYP705A22 has been fused to GFP to test for tissue specificity and timing of expression. Analysis of these fusion proteins will better our understanding of how this gene/protein may be involved in gravitropic signal transduction. In addition, through analysis of microarray data generated from gravistimulated root tips, we were able to identify a root specific P450 in the same family as GPS1. This family has no reported function, and we are planning to characterize this gene as well as to determine if the role of P450s is constitutive in gravitropism or organ specific. We also performed a transcription profiling experiment with gps1. We analyzed RNA extracted from wild-type and gps1 plants 5 minutes after cold gravistimulation and after cold treatment alone. Analysis of these data gave us insight into additional genes that were indirectly affected by the gps1 mutation. We identified 44 genes specific to the gravity stimulus that were not expressed in the cold treatment alone and 76 that were differentially expressed between wild type and gps1 after cold gravistimulation. Of these, we have selected seven and ten genes, respectively, based on the level of expression for further characterization. These include a variety of transcription factors, membrane associated proteins, kinases, and calcium binding proteins among others and should provide us with further insights into the mechanisms of gravitropic signal transduction. The T-DNA insertion in gps3 disrupts a non-coding region on chromosome 1. The region is in the 3’ untranslated region between two genes, an auxin response factor 9 (ARF9) – like gene and a hypothetical gene. We have now determined that the ARF9 like gene is actually the gene disrupted in gps3. We have used a genomic clone of the gene with its endogenous promoter to complement function in the mutant, and using transient expression of a GPS3::GFP fusion have confirmed a nuclear localization of the gene product. We have also built fusions between GFP and GPS3 under control of the endogenous promoter. Those constructs have been transformed into plants and selection of transformants is underway. These plants will be used to determine tissue specificity and temporal expression of the protein. The gene defective in gps2 has now been cloned as well. GPS2 is At5g11150 and encodes a synaptobrevin/ vesicle associated protein that is postulated to be involved in vesicle trafficking. Additional alleles are being evaluated for defects in the gene. Constructs are currently being made to complement the mutant and do temporal and spatial assessment of gene expression.

	And finally, through a continuation of the gps mutant screen, we have identified 12 putative new gps mutants to date and the screen continues. The new mutants fall roughly into the three phenotypes already identified, but preliminary data indicate that a few may represent new allelic groups. The screen and physiological characterization of the new mutants is currently underway. Once complementation groups have been determined, new mutants that do not fall into the complementation groups previously identified will be selected for molecular characterization. Based on these data we will select 1-2 of the most promising new mutants for continued characterization.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
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